Team Involvement in Demonstration

Date:	21 October 2026	**Team ID:**	Piyush-Balodiya-GenAI-2026
Project Name:	Multimodal Construction Intelligence Platform (CONSTRUCTAI)	**Maximum Marks:**	1 Mark

Team Involvement in Demonstration (Part 1 - Solo Lead Execution):

This document records the active participation and end-to-end responsibilities of the solo candidate during the project demonstration. As an individual contributor project, all architectural design, AI vision pipelines, full-stack implementation, and live technical evaluations were single-handedly delivered by Piyush Balodiya.

S. No	Team Member Name	Role in Demo	Section Presented	Contribution Summary	Participation
1	**Piyush Balodiya** Solo Lead	**Project Lead & Architect**	Project Vision, Industry Problem & Architecture	Articulated multi-acre site hazard blind spots, OSHA penalty impacts, and defended the FastAPI + Groq Cloud LPU architecture.	**Active (100% Lead)**
2	**Piyush Balodiya** Solo Lead	**Computer Vision Lead**	Live Demo: PPE Hazard Detection & Audio Sirens	Demonstrated real-time drone snapshot ingestion, sub-900ms missing helmet/vest bounding boxes, and triggered safety warning sirens.	**Active (100% Lead)**
3	**Piyush Balodiya** Solo Lead	**Structural AI Engineer**	Live Demo: IS 456 Crack Diagnostics	Showcased image upload of beam/wall cracks, width classification against IS 456 limits, and automated chemical epoxy repair prescriptions.	**Active (100% Lead)**

Team Involvement in Demonstration

Team Involvement in Demonstration (Part 2 - Full-Stack & Delivery):

Demonstration coverage across material planning, automated compliance reporting, and technical evaluation defense.

S. No	Team Member Name	Role in Demo	Section Presented	Contribution Summary	Participation
4	**Piyush Balodiya** Solo Lead	**Full-Stack Developer**	Live Demo: Weather Radar & BOQ Takeoff	Demonstrated Open-Meteo live rainfall threshold tracking to prevent pour washouts and generated instant cement/rebar quantity estimates.	**Active (100% Lead)**
5	**Piyush Balodiya** Solo Lead	**Backend & Reports Lead**	Live Demo: Executive PDF Engine	Executed automated 1-click ReportLab PDF generation, verified zero server image retention, and inspected tamper-proof audit trails.	**Active (100% Lead)**
6	**Piyush Balodiya** Solo Lead	**Solo Technical Presenter**	Q&A;, Defense & Roadmap Presentation	Successfully defended latency benchmarks, offline SQLite fallback mode, and outlined drone edge roadmap to project evaluators.	**Active (100% Lead)**

Team Coordination Notes (Solo Project Execution):

Comprehensive record of project ownership, self-managed coordination, and technical risk resolution.

Aspect	Details & Solo Execution Protocol
Team Leader / Coordinator	**Piyush Balodiya** (Sole Developer, End-to-End System Architect & Project Owner)
Overall Team Coordination Rating (1-5)	**5 / 5 (Flawless unified delivery, zero inter-member dependency delays, 100% individual accountability)**
Any issues during demo	Potential Groq Cloud API rate-limiting and temporary remote network dropouts during live high-resolution image uploads.
How issues were resolved	Engineered offline-first local SQLite caching for instant IS 456 remediation guidelines, with pre-loaded mock telemetry fallbacks ensuring zero presentation interruption.